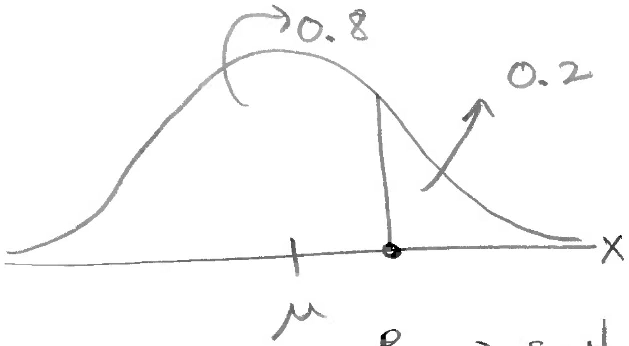
Example: Calculating percentiles with the normal distribution.

Consider $Z = \frac{X - \mu}{\sigma}$, thus $X = \mu + Z\sigma$.

Percentiles or quartiles from the normal distribution can be obtained using the formula.

e.g.



$P_{80} \Rightarrow$ 80th percentile.

Let $X \sim N(\mu = 18, \sigma^2 = 25)$ $\sigma = 5$ (X = download time)

Calculate:

(a) The 80th percentile of download time.

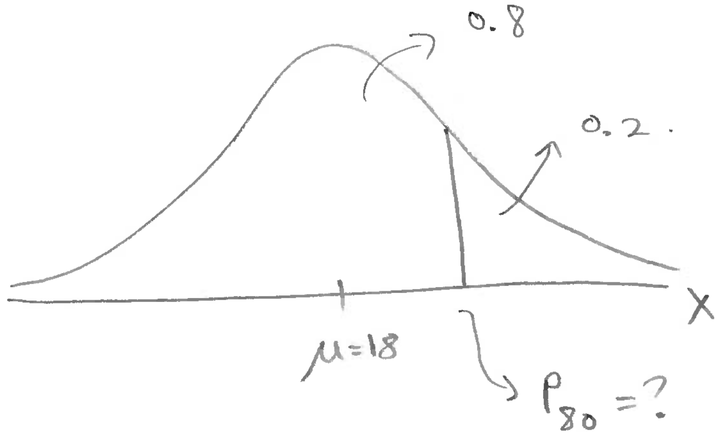
(b) Find X such that 5% of download times is less than X.

(c) Find Q_1, Q_2 and Q_3 (the quartiles).

(d) Given $P(a < X < b) = 0.9$, find a and b by assuming they are symmetric around μ.

Solutions:

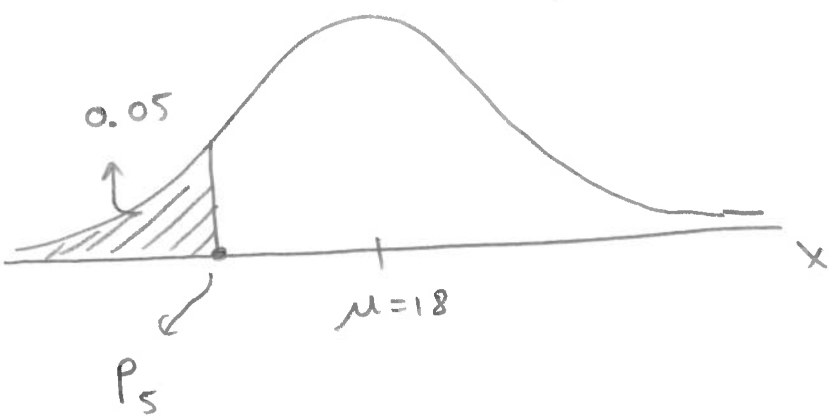
(a)



$$P_{80} = \mu + Z_{80}\,\sigma$$

$$= 18 + (0.8416)(5) = 22.2081$$

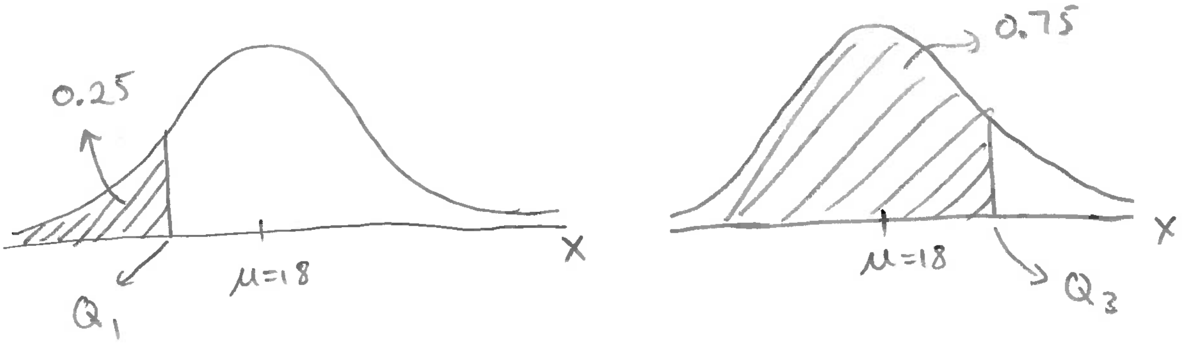
(b)



$$P_5 = \mu + Z_5\,\sigma$$

$$= 18 + (-1.6449)(5) = 9.7757$$

(c) We know $Q_2 = \mu = 18$ (median = mean)



Thus,

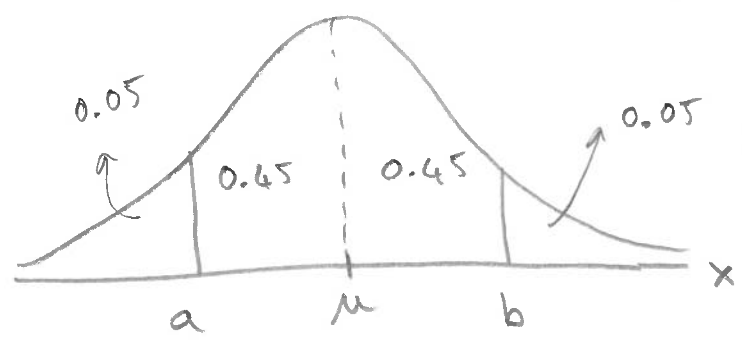
$$Q_1 = P_{25} = \mu + Z_{25}\,\sigma$$

$$= 18 + (-0.6745)(5) = \underline{14.6276}$$

$$Q_3 = P_{75} = \mu + Z_{75}\,\sigma$$

$$= 18 + (0.6745)(5) = \underline{21.3724}$$

(d)



$$P(a < X < b) = 0.9$$

$$a \Rightarrow P_5 = \mu + Z_5\,\sigma$$

$$= 18 + (-1.6449)(5) = \underline{9.7757}$$

$$b \Rightarrow P_{95} = \mu + Z_{95}\,\sigma$$

$$= 18 + (1.6449)(5) = \underline{26.2243}$$